

UNIVERSITY of ASIA PACIFIC

Department of Computer Science & Engineering (CSE)

Course Title: Database Systems Lab

Course Code: CSE - 212

Project Name: Filling Station Management System

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Table Of Content

1. Project Description 3
2. Entities (Tables) 4
3. Table of Work 5

4. Relationship Between Entities 8

5. ER- Diagram 10
6. Schema- Diagram 11
7. SQL Queries 12
8. CEP Mapping 22
8. Conclusion 27

1.Project Description

The Filling Station Management System is a relational database project developed to improve fuel station operations and provide customers with convenient access to fuel-related information. Fuel stations often face challenges such as fluctuating fuel prices, manual sales and inventory management, limited information sharing, and difficulties in monitoring fuel availability. These issues can lead to operational inefficiencies, data inaccuracies, and unnecessary waiting times for customers.

This project leverages MySQL to implement a structured relational database that supports centralized fuel management and networking among multiple fuel stations. The system is designed to manage essential information, including fuel types, prices, storage tanks, stock levels, sales records, and station details. Primary and foreign key constraints help maintain data consistency and ensure accurate relationships between related entities.

The primary goal is to offer a comprehensive database solution that can:

1. Provide updated fuel prices and availability information to help customers find suitable nearby fuel stations.
2. Enable customers to compare fuel prices and make informed decisions based on available information.
3. Centralize the management of fuel sales, inventory, and station information through a unified platform.
4. Automate sales recording and stock tracking to reduce manual errors and improve operational efficiency.
5. Monitor fuel inventory levels and identify when replenishment is required.
6. Generate reports and analyze sales trends, fuel demand, and overall business performance.
7. Connect multiple fuel stations to facilitate information sharing, improve coordination, and enhance customer accessibility.

By implementing a structured SQL-based database, this system aims to simplify daily fuel station operations, improve price transparency, and support data-driven business decisions. Its centralized approach provides a foundation for better coordination between fuel stations and more efficient service for customers. In the future, the database can be integrated with a web or mobile application to provide real-time fuel price updates, station availability, and additional customer services.

2.Entities(Tables)

1. **Fuel**
   - Fuel_Id(Primary Key)
   - Fuel_Name
   - Price_per_Leter
   - Status

2. Storage_Tank
   - Tank_Id(Primary Key)
   - Fuel_Id(Foreign Key)
   - Capacity
   - Stock
   - Location
   - Pump_Id

3. Vehicle
   - Vehicle_Id(Primary Key)
   - Vehicle_type
   - Fuel_Id(Foreign Key)
   - Customer_Id(Foreign Key)

4. Customer
   - Customer_Id(Primary Key)
   - Customer_Name
   - Customer_Phone

5. Sales_Transaction
   - Sale_Id(Primary Key)
   - Customer_Id(Foreign Key)
   - Sale_Date
   - Quantity
   - Total_amount

6. Payment
   - Payment_Id(Primary Key)
   - Sale_Id(Foreign Key)
   - Payment_Type
   - Status

7. Subscribers
   - Subscriber_Id(Primary Key)
   - Customer_Id(Foreign Key)
   - Subs_Id(Foreign Key)
   - Fuel_Id(Foreign Key)
   - Quantity

8. Subscription_Plan
   - Subs_Id(Primary Key)
   - Subs_Plan
   - Sub_Point_Id(Foreign Key)
   - Duration_Months

9. Sub_Point
   - Sub_Point_Id(Primary Key)
   - Sub_Point
   - Facility

3.Table of Work

1. **Payment:**
   - **a)** Payment_id (Primary Key)
   - **b)** Sale_id (Foreign Key)
   - **c)** Payment_Type
   - **d)** Status

This table stores payment information for each sales transaction, including the payment type and payment status. Here, Sale_id is a foreign key that references the Sales_transaction table.

2. **Sales_transaction:**
   - **a)** Sale_id (Primary Key)
   - **b)** Customer_id (Foreign Key)
   - **c)** Sale_date
   - **d)** Quantity
   - **e)** Total_amount

This table records fuel sales transactions, including the customer details, sale date, quantity of fuel purchased, and total amount. Here, Customer_id is a foreign key that references the Customer table.

3. **Customer:**
   - **a)** Customer_id (Primary Key)
   - **b)** Customer_Name
   - **c)** Customer_phone_no
   - **d)** Vehicle_id (Foreign Key)

This table stores customer information, including names, phone numbers, and associated vehicles. Here, Vehicle_id is a foreign key that references the Vehicle table.

4. **Vehicle:**

   - **a)** Vehicle_id (Primary Key)
   - **b)** Vehicle_type
   - **c)** Fuel_id (Foreign Key)

This table maintains information about customer vehicles, including their types and the fuel they use. Here, Fuel_id is a foreign key that references the Fuel table.

5. **Fuel:**

    a) Fuel_id (Primary Key)
    b) Fuel_Name
    c) Price_per_liter
    d) Status

This table stores information about different fuel types, their prices per liter, and their current status. It helps maintain fuel price information and supports fuel availability management.

6. **StorageTank:**

    a) Tank_Id (Primary Key)
    b) Fuel_Id (Foreign Key)
    c) Capacity
    d) Stock
    e) Location
    f) Pump_Id

This table monitors fuel storage tanks, including their capacity, current stock, location, and associated pump ID. Here, Fuel_Id is a foreign key that references the Fuel table. It helps track fuel inventory and identify when replenishment is needed.

7. **Subscribers:**

    a) Subscriber_id (Primary Key)
    b) Customer_id (Foreign Key)
    c) Subs_id (Foreign Key)
    d) Fuel_id (Foreign Key)
    e) Vehicle_id (Foreign Key)
    f) Quantity

This table stores customer subscription information, including the selected subscription plan, fuel type, associated vehicle, and fuel quantity. Customer_id, Subs_id, Fuel_id, and Vehicle_id are foreign keys that reference the Customer, Subscription_Plan, Fuel, and Vehicle tables, respectively.

8. **Subscription_Plan:**

    a) Subs_id (Primary Key)
    b) Subs_Plan
    c) Sub_Point_id (Foreign Key)
    d) Duration_Months

This table stores different subscription plans, their names, associated subscription points, and durations in months. Here, Sub_Point_id is a foreign key that references the Sub_Point table.

9. **Sub_Point:**

    a) Sub_Point_id (Primary Key)
    b) Sub_Point
    c) Facility

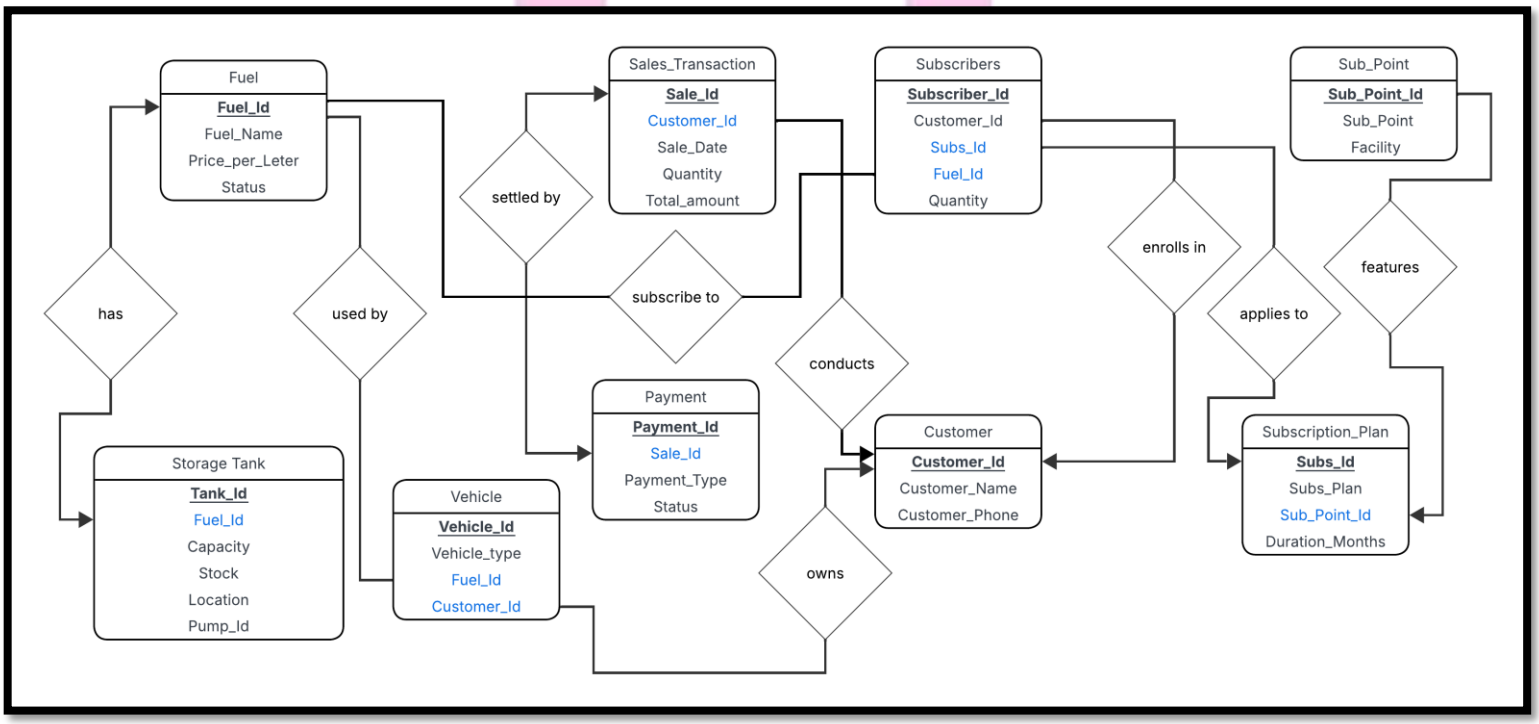
This table stores information about subscription points and their available facilities. It helps customers identify the facilities associated with different subscription plans.

4. Relationship Between Entities

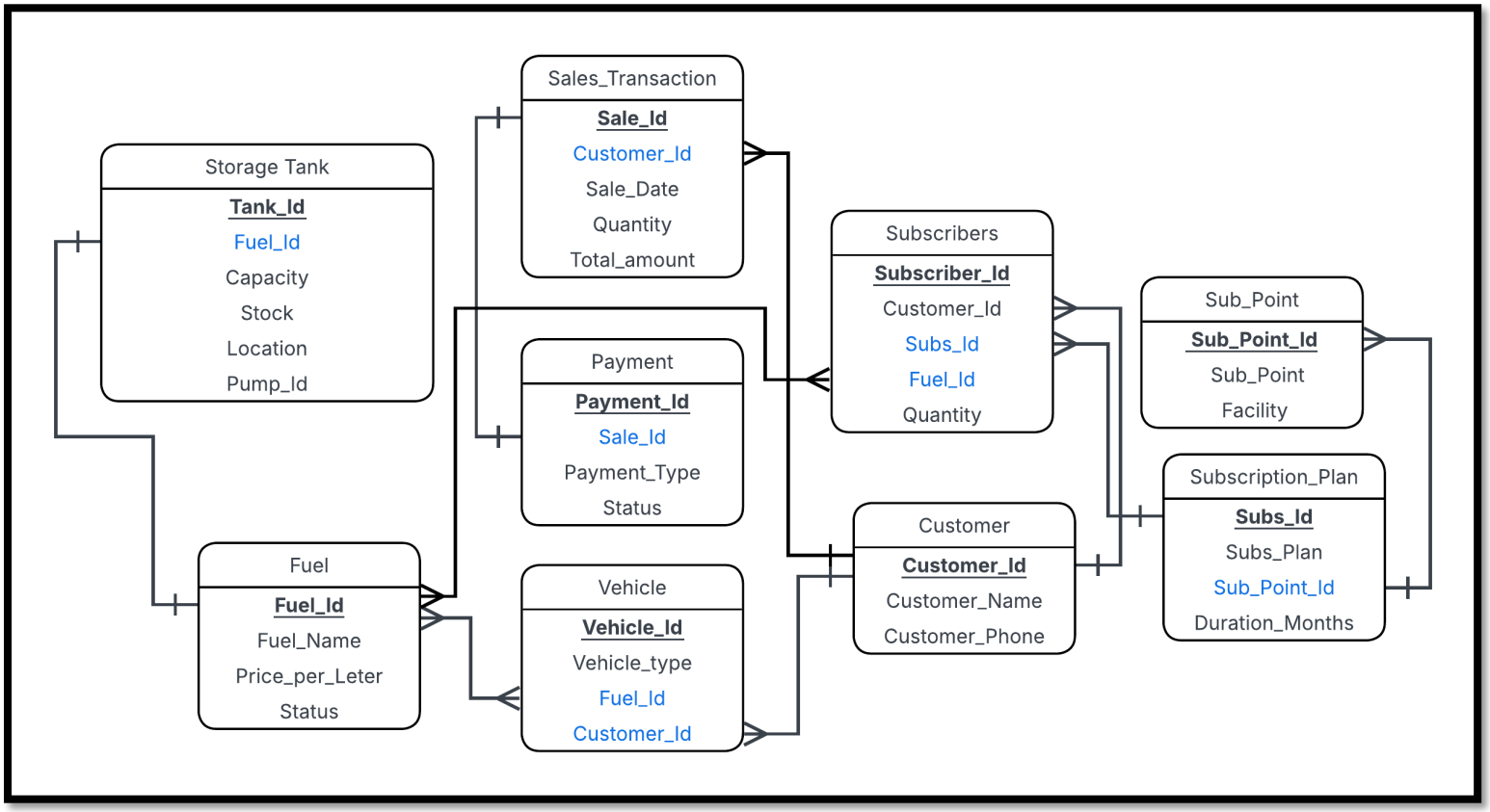
- **Customer ↔ Vehicle (owns)**
    - **Cardinality:** 1: M
    - **Description:** A customer can own multiple vehicles; each vehicle belongs to one customer.
- **Customer ↔ Sales_Transaction (conducts)**
    - **Cardinality:** 1: M
    - **Description:** A customer can make multiple transactions; each transaction belongs to one customer.
- **Customer ↔ Subscribers (enrolls in)**
    - **Cardinality:** 1: M
    - **Description:** A customer can have multiple subscription records; each record belongs to one customer.

- **Subscription_Plan ↔ Subscribers (applies to)**
  - **Cardinality:** 1: M
  - **Description:** A plan can apply to multiple subscribers; each subscriber uses one plan.
- **Sales_Transaction ↔ Payment (settled by)**
  - **Cardinality:** 1: 1
  - **Description:** Each transaction has one payment, and each payment settles one transaction.
- **Fuel ↔ Storage_Tank (has)**
  - **Cardinality:** 1: 1
  - **Description:** Each fuel type is linked to one storage tank, and vice versa.
- **Sub_Point ↔ Subscription_Plan (features)**
  - **Cardinality:** M: 1
  - **Description:** Multiple sub-points feature one subscription plan.
- **Fuel ↔ Vehicle (used by):**
  - **Cardinality:** M: N
  - **Description:** A fuel type is used by many vehicles, and vehicles can use multiple fuels.
- **Fuel ↔ Subscribers (subscribe to)**
  - **Cardinality:** M: N
  - **Description:** A fuel type can have many subscribers, and subscribers can link to multiple fuels.

5.ER Diagram


Fuel
Fuel_Id
Fuel_Name
Price_per_Leter
Status
Sales_Transaction
Sale_Id
Customer_Id
Sale_Date
Quantity
Total_amount
Subscribers
Subscriber_Id
Customer_Id
Subs_Id
Fuel_Id
Quantity
Sub_Point
Sub_Point_Id
Sub_Point
Facility
settled by
has
used by
subscribe to
conducts
enrolls in
applies to
features
Payment
Payment_Id
Sale_Id
Payment_Type
Status
Customer
Customer_Id
Customer_Name
Customer_Phone
Subscription_Plan
Subs_Id
Subs_Plan
Sub_Point_Id
Duration_Months
Storage Tank
Tank_Id
Fuel_Id
Capacity
Stock
Location
Pump_Id
Vehicle
Vehicle_Id
Vehicle_type
Fuel_Id
Customer_Id
owns

6. Schema Diagram



7. SQl Queries

```sql
-- 1. WHERE - Find Available Fuels
-- Available Fuel গুলো দেখাবে
SELECT * FROM Fuel
WHERE Status = 'available';
```

Fuel_Id	Fuel_Name	Price_per_liter	Status
1	Octen	145.00	available
2	Diesel	115.00	available
3	LPG	70.40	available

```sql
-- 2. ORDER BY - Sort Fuel Prices
-- Fuel-এর দাম ছোট থেকে বড় সাজাবে
SELECT * FROM Fuel
ORDER BY Price_per_liter ASC;
```

Fuel_Id	Fuel_Name	Price_per_liter	Status
3	LPG	70.40	available
2	Diesel	115.00	available
5	Petrol / Gasoline	140.00	not available
1	Octen	145.00	available
4	Lubricant	450.00	not available

```sql
-- 3. LIKE - Search Customer by Name
-- যেসব Customer-এর নাম T দিয়ে শুরু তাদের দেখাবে
SELECT * FROM Customer
WHERE Customer_Name LIKE 'T%';
```

```
+-------------+---------------+--------------------+
| Customer_Id | Customer_Name | Customer_phone_no |
+-------------+---------------+--------------------+
|           1 | Tanvir Ahmed  | 01712-458921       |
|          18 | Tania Sultana | 01955-778899       |
+-------------+---------------+--------------------+
```

```
-- 4. BETWEEN - Find Sales Within Price Range
-- ৩০০০ থেকে ৭০০০ টাকার মধ্যে Sales দেখাবে
SELECT * FROM Sales_transaction
WHERE Total_amount BETWEEN 3000 AND 7000;
```

```
+---------+-------------+------------+----------+--------------+
| Sale_Id | Customer_Id | Sale_date  | Quantity | Total_amount |
+---------+-------------+------------+----------+--------------+
|    1001 |           1 | 2026-01-10 |    45.00 |      6525.00 |
|    1002 |           2 | 2026-01-12 |    32.50 |      3737.50 |
|    1004 |           4 | 2026-01-18 |    60.00 |      6900.00 |
|    1006 |           6 | 2026-01-22 |    40.00 |      4600.00 |
|    1009 |           9 | 2026-02-02 |    30.00 |      4350.00 |
|    1010 |          10 | 2026-02-05 |    50.00 |      3520.00 |
|    1011 |          11 | 2026-02-08 |    35.00 |      5075.00 |
|    1014 |          14 | 2026-02-17 |    55.00 |      6325.00 |
|    1016 |          16 | 2026-02-23 |    22.00 |      3190.00 |
|    1017 |          17 | 2026-02-26 |    45.00 |      5175.00 |
|    1020 |          20 | 2026-03-07 |    38.00 |      5510.00 |
|    1021 |          21 | 2026-03-10 |    42.00 |      4830.00 |
|    1022 |          22 | 2026-03-13 |    28.00 |      3220.00 |
|    1023 |          23 | 2026-03-16 |    50.00 |      5750.00 |
|    1024 |          24 | 2026-03-19 |    33.00 |      4620.00 |
|    1027 |          27 | 2026-03-28 |    48.00 |      5520.00 |
|    1030 |          30 | 2026-04-05 |    40.00 |      4600.00 |
+---------+-------------+------------+----------+--------------+
```

```sql
-- 5. OR - Find Pre-Paid or Pending Payments
-- Pre-Paid অথবা Pending Payment দেখাবে
SELECT * FROM Payment
WHERE Payment_Type = 'Pre-Paid'
OR Status = 'Pending';
```

Payment_Id	Sale_Id	Payment_Type	Status
5001	1001	Pre-Paid	Paid
5003	1003	Pre-Paid	Paid
5005	1005	Pre-Paid	Pending
5007	1007	Pre-Paid	Paid
5009	1009	Pre-Paid	Paid
5010	1010	Post-Paid	Pending
5011	1011	Pre-Paid	Paid
5013	1013	Pre-Paid	Paid
5015	1015	Pre-Paid	Pending
5017	1017	Pre-Paid	Paid
5018	1018	Pre-Paid	Paid
5019	1019	Pre-Paid	Paid
5020	1020	Post-Paid	Pending
5021	1021	Pre-Paid	Paid
5023	1023	Pre-Paid	Paid
5025	1025	Pre-Paid	Paid
5026	1026	Post-Paid	Pending
5027	1027	Pre-Paid	Paid
5029	1029	Pre-Paid	Paid

```sql
-- 6. COUNT - Count Total Customers and Calculate Total Sales
-- মোট Customer সংখ্যা দেখাবে and সব Sales-এর মোট Amount দেখাবে
SELECT
    C.Total_Customers,
    S.Total_Sales
FROM
    (SELECT COUNT(*) AS Total_Customers
     FROM Customer) AS C
CROSS JOIN
    (SELECT SUM(Total_amount) AS Total_Sales
     FROM Sales_transaction) AS S;
```

Total_Customers	Total_Sales
30	149745.40

```sql
-- 7. HAVING - Find Fuel Used by More Than 5 Vehicles
-- ৫টির বেশি Vehicle-এ ব্যবহৃত Fuel দেখাবে
SELECT F.Fuel_Id, F.Fuel_Name, COUNT(V.Vehicle_Id) AS Total_Vehicles
FROM Vehicle V
INNER JOIN Fuel F ON V.Fuel_Id = F.Fuel_Id
GROUP BY
    F.Fuel_Id,
    F.Fuel_Name;
```

Fuel_Id	Fuel_Name	Total_Vehicles
1	Octen	9
2	Diesel	13
3	LPG	4
5	Petrol / Gasoline	4

```sql
-- 8. CASE WHEN - Categorize Fuel Prices
-- Fuel-এর দাম অনুযায়ী Category তৈরি করবে
SELECT Fuel_Name, Price_per_liter,
CASE
    WHEN Price_per_liter < 100 THEN 'Cheap'
    WHEN Price_per_liter <= 200 THEN 'Medium'
    ELSE 'Expensive'
END AS Price_Category
FROM Fuel;
```

Fuel_Name	Price_per_liter	Price_Category
Octen	145.00	Medium
Diesel	115.00	Medium
LPG	70.40	Cheap
Lubricant	450.00	Expensive
Petrol / Gasoline	140.00	Medium

```sql
-- 9. INNER JOIN - Customer and Vehicle Information
-- Customer-এর নাম ও Vehicle-এর তথ্য দেখাবে
SELECT Customer.Customer_Name,
       Vehicle.Vehicle_type
FROM Customer
RIGHT JOIN Vehicle
ON Customer.Customer_Id = Vehicle.Customer_Id;
```

Customer_Name	Vehicle_type
Tanvir Ahmed	Bike
Tanvir Ahmed	Car
Tanvir Ahmed	Bus
Fariha Tabassum	Truck
Fariha Tabassum	Microbus
Shahriar Hossain	CNG Auto Rickshaw
Shahriar Hossain	Pickup
Nusrat Jahan	Van
Rakib Hasan	Jeep
Sadia Afrin	Motorcycle
Imtiaz Mahmud	SUV
Lamia Islam	Covered Van
Zubayer Rahman	Minibus
Farzana Yasmin	Taxi Cab
Mahbub Alam	Electric Hybrid Car
Anika Tasnim	Sports Car
Shariful Islam	Heavy Truck
Priya Sarker	Delivery Scooter
Sabbir Hossain	Private Coach
Mumu Akter	Ambulance
Rashed Khan	Fire Service Truck
Tania Sultana	Police Patrol Car
Al-Amin Mia	Micro Truck
Bristy Roy	Three Wheeler
Fahim Faisal	Tour Bus
Nadia Chowdhury	Cargo Lorry
Sumon Karmakar	Executive Sedan
Mehedi Hasan	Cruiser Bike
Nazmul Haque	City Bus
Sabrina Yesmin	Postal Van

```sql
-- 11. MULTIPLE JOIN - Customer, Sales and Payment
-- Customer-এর নাম, Sales Amount ও Payment Status দেখাবে
SELECT Customer.Customer_Name,
       Sales_transaction.Total_amount,
       Payment.Payment_Type,
       Payment.Status
FROM Customer
INNER JOIN Sales_transaction
ON Customer.Customer_Id = Sales_transaction.Customer_Id
INNER JOIN Payment
ON Sales_transaction.Sale_Id = Payment.Sale_Id;
```

Customer_Name	Total_amount	Payment_Type	Status
Tanvir Ahmed	6525.00	Pre-Paid	Paid
Fariha Tabassum	3737.50	Post-Paid	Paid
Shahriar Hossain	1408.00	Pre-Paid	Paid
Nusrat Jahan	6900.00	Post-Paid	Paid
Rakib Hasan	2247.50	Pre-Paid	Pending
Sadia Afrin	4600.00	Post-Paid	Paid
Imtiaz Mahmud	2875.00	Pre-Paid	Paid
Lamia Islam	8625.00	Post-Paid	Paid
Zubayer Rahman	4350.00	Pre-Paid	Paid
Farzana Yasmin	3520.00	Post-Paid	Pending
Mahbub Alam	5075.00	Pre-Paid	Paid
Anika Tasnim	10350.00	Post-Paid	Paid
Shariful Islam	1740.00	Pre-Paid	Paid
Priya Sarker	6325.00	Post-Paid	Paid
Sabbir Hossain	9200.00	Pre-Paid	Pending
Mumu Akter	3190.00	Post-Paid	Paid
Rashed Khan	5175.00	Pre-Paid	Paid
Tania Sultana	1302.40	Pre-Paid	Paid
Al-Amin Mia	7475.00	Pre-Paid	Paid
Bristy Roy	5510.00	Post-Paid	Pending
Fahim Faisal	4830.00	Pre-Paid	Paid
Nadia Chowdhury	3220.00	Post-Paid	Paid
Sumon Karmakar	5750.00	Pre-Paid	Paid
Mehedi Hasan	4620.00	Post-Paid	Paid
Nazmul Haque	8050.00	Pre-Paid	Paid
Sabrina Yesmin	1450.00	Post-Paid	Pending
Juel Rana	5520.00	Pre-Paid	Paid
Moniruzzaman	2875.00	Post-Paid	Paid
Rumana Khatun	8700.00	Pre-Paid	Paid
Asif Akbar	4600.00	Post-Paid	Paid

```sql
-- 12. SUBQUERY - Find Fuels Above Average Price
-- গড় দামের চেয়ে বেশি দামের Fuel দেখাবে
SELECT Fuel_Name, Price_per_liter
FROM Fuel
WHERE Price_per_liter >
(
    SELECT AVG(Price_per_liter)
    FROM Fuel
);
```

```
+-----------+-----------------+
| Fuel_Name | Price_per_liter |
+-----------+-----------------+
| Lubricant |          450.00 |
+-----------+-----------------+
```

```sql
-- 13. GROUP BY + HAVING + JOIN + ORDER BY
-- যেসব Customer-এর মোট Sales ৫০০০ টাকার বেশি তাদের দেখাবে
SELECT Customer.Customer_Name,
       SUM(Sales_transaction.Total_amount) AS Total_Sales
FROM Customer
INNER JOIN Sales_transaction
ON Customer.Customer_Id = Sales_transaction.Customer_Id
GROUP BY Customer.Customer_Id, Customer.Customer_Name
HAVING SUM(Sales_transaction.Total_amount) > 5000
ORDER BY Total_Sales DESC;
```

```
+---------------+-------------+
| Customer_Name | Total_Sales |
+---------------+-------------+
| Anika Tasnim  |    10350.00 |
| Sabbir Hossain |     9200.00 |
| Rumana Khatun |     8700.00 |
| Lamia Islam   |     8625.00 |
| Nazmul Haque  |     8050.00 |
| Al-Amin Mia   |     7475.00 |
| Nusrat Jahan  |     6900.00 |
| Tanvir Ahmed  |     6525.00 |
| Priya Sarker  |     6325.00 |
| Sumon Karmakar |     5750.00 |
| Juel Rana     |     5520.00 |
| Bristy Roy    |     5510.00 |
| Rashed Khan   |     5175.00 |
| Mahbub Alam   |     5075.00 |
+---------------+-------------+
```

8.CEP Mapping:

1. Knowledge Profiles(K's) are addressed through the project and mapping among K's, CO's, and PO's:-

K's	Attributes	How K's are Addressed in this project	CO	PO
K2	Mathematics	Numerical analysis and calculations are applied to compute total sales amounts, fuel quantities, storage tank capacities, and stock depletion rates.	CO2, CO4	PO1, PO3, PO6, PO7, PO8
K3	Engineering Fundamentals	Fundamental database principles are applied to design and operate the fuel station database, including the formulation of primary/foreign key relationships, normalization, and integrity constraints to ensure proper data management.	CO1, CO2, CO3	PO1
K5	Engineering Design	The project involves designing a structured ER diagram and relational schema diagram, effectively representing data flow and relationships	CO3, CO4	PO3, PO5

		among entities like fuel types, storage tanks, vehicles, sales transactions, and customer subscriptions.		
K6	Engineering Practice	The database design is fully implemented using MySQL. Tables are created using DDL statements, data is populated, and complex queries (joins, aggregations, subqueries) are executed using DML via MySQL Workbench.	CO3, CO5	PO1, PO3, PO5

2. Complex Engineering Problem solving (P's) are addressed through the project and mapping among K's, CO's, and PO's:-

P's	Attributes	How P's are addressed through this project	CO's	PO's
P1	Depth of knowledge required	The project requires studying core database management system design principles (K_3), creating comprehensive ER and schema diagrams (K_5), and executing practical backend implementations (K_6) tailored to fuel retail	CO3, CO4, CO5	PO1, PO2, PO3, PO7

		operations.		
P3	Depth of analysis required	Functional queries, multi-table joins, and aggregations are developed using abstract thinking and in-depth analysis to track fuel stock thresholds, monitor customer spending patterns, and evaluate payment statuses.	CO3, CO5	PO3
P6	Extent of stake	The system is designed to accommodate the unique data requirements of diverse stakeholders—such as vehicle owners/customers, station operators, and management—maintaining smooth record-keeping and operational transparency.	CO3, CO5	PO2, PO7
P7	Interdependence	High levels of data interdependence are demonstrated by components linking Storage_Tank to Fuel, Sales_Transaction to Customer and Payment, and	CO3, CO4, CO5	PO2, PO7

		Subscribers to Subscription_Plan and Vehicle, guaranteeing proper relational connectivity and reporting integrity.		

3. Complex Engineering Activity(A's) are addressed through the project and mapping among K's, CO's, and PO's:-

A's	Attributes	How A's are addressed through our project	CO's	PO's
A1	Range of resources	The project utilizes diverse operational data resources, including fuel pricing catalogs, storage tank capacity logs, vehicle registries, payment statuses, and subscription records to manage station operations effectively.	CO1, CO2, CO8	PO1, PO3, PO5, PO6, PO7, PO8
A2	Level of interaction	Complex interactions between system entities and users are modeled accurately, such as customers purchasing fuel via Sales_Transaction, linking vehicles to specific fuel types, and managing pre-paid or post-paid Payment	CO7, CO8, CO9	PO1, PO3, PO5, PO6

		settlements.		
A3	Innovation	Innovations include utilizing advanced SQL features such as case-when categorization for fuel pricing, cross-join aggregations for total sales computation, and multi-table joins to automate inventory tracking, payment verification, and customer subscription management.	CO3, CO5, CO6, CO7	PO3, PO6, PO7, PO8
A5	Familiarity	The project builds on established database concepts like relational models, normalization, and constraints, applying them directly to the real-world context of fuel distribution to minimize data redundancy, track inventory depletion, and ensure accurate financial record-keeping.	CO5, CO6, CO7	PO10, PO11, PO12

9.Conclusion:

The Filling Station Management System developed in this project demonstrates how relational databases can be used to organize, manage, and improve fuel station operations. Through structured SQL tables, well-defined relationships, and meaningful queries, the system supports essential functionalities such as managing fuel sales, monitoring inventory, maintaining customer and station records, processing payments, and organizing subscription plans. It provides a foundation for improving operational efficiency, reducing manual errors, and supporting better decision-making in fuel management.

Advantages:

1. **Efficient Sales Management:** The system helps record and manage fuel sales transactions more quickly and accurately.
2. **Improved Inventory Tracking:** Storage tank records help monitor fuel stock levels and identify when replenishment is required.
3. **Centralized Data Management:** Customer, fuel, and station-related information can be organized in a structured database for easier access and management.
4. **Simplified Payment Processing:** Payment and transaction records make it easier to track payment types, statuses, and sales amounts.
5. **Reduced Manual Work:** Automated database operations and SQL queries help minimize repetitive tasks and reduce the risk of human error.

Limitations:

1. **No User Interface:** The project primarily focuses on the database backend and does not include a complete graphical interface for customers or station administrators.
2. **Limited Real-Time Updates:** Real-time fuel prices and availability require integration with live station data and an appropriate application interface.
3. **Limited Queue Information:** The system cannot accurately estimate waiting times unless queue lengths and service activity are recorded and updated.
4. **Limited Station Networking:** Connecting multiple fuel stations and synchronizing their information requires additional networking functionality and supporting application logic.
5. **No Advanced Security Features:** Authentication, role-based access control, and additional security mechanisms are needed to protect customer information and sensitive business records.

Drive Link:

https://github.com/saadbinharun7/Filling-Station-Management-System---Project-for-DBMS?fbclid=IwY2xjawVcCdtleHRuA2FlbQIxMABwZG9mBWJyaWQRMThtMjFQcFlSWGxQVUdSSVhzcnRjBmFwcF9pZBAyMjIwMzkxNzg4MjAwODkyAAEfBaT3TvkxGJvTkx09r5I4lmUfm3Am4QwzOfGhZRs2UFaXtbwXH0r7EM3VrWw_aem_FSqC3WSF2lvMo-i0QkgoAw

